Index

Note: Page numbers followed *f* indicate figures and *t* indicate tables.

ADVANCED OPERATIVE DENTISTRY: RICKETTS AND BARTLETT

A

Abfraction, 46*f*
  definition, 45
Abrasion
  aetiology, 50
  definition, 45
  molar teeth, 47
Absorption, light, 188
Abutment
  conventional bridges, 240, 240*f*
  definition, 239
Acid microabrasion, 66, 66*f*
Acrylic splints, bruxism, 70
Active posts, 96
Acute pulpitis, 38
Addition cured silicones, 166
Aδ-fibres, dentine pulp complex, 35
Adhesive cements
  cores, 90–91
  gold crowns, 107, 107*f*
  minimal preparation bridges, 229–230, 230*f*
Aesthetic(s), 187–200
  aesthetic analysis, 194–198
  conventional bridges, 240*f*, 241, 241*f*
  custom-formed resin replica crowns, 181–182
  dento-facial relationships, 195–196
    midline, 195, 195*f*
    smile line, 195–196
    vertical dimensions, 196
  dento-gingival relationships, 196–198
    black triangles, 196, 198*f*
    bony factors, 196
    gingival contour, 198, 199*f*
    hard–soft tissue relationship, 198
    hard tissues, 196, 197*f*, 198*f*
    incisal openings, 196
    soft tissues, 197–198
    tooth colour, 196, 198*f*
    tooth size, 196, 198*f*
  facial relationships, 194–195
    facial midline, 194, 195*f*
    Golden proportion, 195, 195*f*, 197*f*
    horizontal lines of face, 195
  horizontal tooth form, 194
  metal–ceramic crowns, 132, 132*f*
  porcelain jacket crowns, 142, 142*f*
  problems *see below*
  provisional restorations, 178
  vertical tooth form, 194
Aesthetic problems, 55–68
  black triangles, 59, 59*f*
  causes, 55–62
  malformed teeth, 61
    peg laterals, 61
    trauma, 61
  missing lateral incisors, 62
  missing teeth replacement, 215, 216*f*
  preoperative assessment, 62–63
    colour, 62–63
    diagnostic wax-up, 62, 62*f*
    patient happiness, 62
  shape and size, 56–59, 56*f*
    gingival contours, 55–56, 56*f*
  teeth colour, 56–59
    amelogenesis imperfecta, 58–59, 59*f*
    canine, 56–57, 57*f*
    dentinogenesis imperfecta, 58, 58*f*
    fluorosis, 57, 57*f*
    genetic disorders, 58
    preoperative assessment, 62–63
    tetracycline staining, 57, 57*f*
  tooth spacing, 59–60
    diastemas, 59, 59*f*
    restorative intervention, 59–60, 59*f*, 60*f*
Aesthetic zone, gingival management (retraction), 172
Agar, impression materials, 163–164
Aggressive periodontitis, 25*f*, 26–27, 27*f*
Alcohol consumption, periodontal disease, 19, 28
Alginate *see* Impression materials
All-ceramic crowns, 141–150
  CAD-CAM, 147–149
    Cerec machine, 148
    leucite-reinforced ceramic, 149
    Nobel Biocare, 147, 147*f*
    procera crowns, 147
    3M ESPE, 147, 148*f*
    VITA In-Ceram, 147
    Z100 composite, 149
  castable glass ceramic, 143*f*, 144–145, 145*t*, 146*f*
  cementation, 149
  coping materials, 146, 147*f*
  decreasing tooth tissue reduction, 143–144
  dentine-bonded crowns, 143–144, 143*f*
    fitting surface, 144
    fragility, 144
    manufacture, 144
    shoulders, 143, 143*f*, 144*f*
  generic crown preparations, 146, 146*f*
  glass ceramic materials, 144–147, 145*t*
  glass-infused ceramic copings, 146–147

All-ceramic crowns (*Continued*)
hot-pressed ceramics, 145–146, 145*t*, 146*f*
injection-moulded ceramics, 145–146, 145*t*, 146*f*
machining, 145*t*
see also specific crowns
Alveolar bone loss, severe, 221, 222*f*
Amalgam, cores *see* Cores
Amelogenesis imperfecta, teeth colour, 58–59, 59*f*
Anterior guidance
gold crowns, 105
mandible excursive movement, 74–76
Anterior teeth
attrition, 49*f*
cores *see* Cores
erosion, 48*f*
metal–ceramic crowns *see* Metal–ceramic crowns
minimal preparation bridges *see* Minimal preparation bridges
tooth wear, 46–47, 46*f*, 47*f*, 48*f*
see also Canine teeth; Incisor teeth
Ante's law, conventional bridges, 249
Appliances, periodontal disease, 20
Approximal preparation
anterior metal–ceramic crowns, 139, 139*f*
gold crown axial reduction, 112, 112*f*, 113*f*
posterior metal–ceramic crowns, 136–137, 136*f*, 137*f*
Articulators, 77–84
average value, 77, 77*f*
fully adjustable, 78
semi-adjustable, 78, 78*f*
arcon *vs.* non-arcon, 78
facebows, 78–79, 78*f*, 79*f*, 80*f*
interocclusal record, 79–82, 81*f*, 82*f*, 83*f*, 84*f*
model mounting, 78–82
simple hinge, 77, 77*f*
Artificial saliva, dental caries, 9–10
Attachment loss, periodontal disease assessment, 23, 23*f*
Attrition
aetiology, 50
anterior teeth, 49*f*
definition, 45
molar teeth, 47–49, 48*f*
Average value articulators, 77, 77*f*
Awake bruxism, 69–70
Axial reduction
gold crowns *see* Gold crowns
metal–ceramic crowns, 134

B

Bacterial penetration
dentine pulp complex, 37–38
provisional restorations, 178
Base metal alloys, metal–ceramic crowns, 130
Basic periodontal examination (BPE), 20–21, 20*t*, 22*t*
Bennett angle, 74, 75*f*
Biocompatibility, gold alloys, 117
Biofilm removal, dental caries diagnosis, 3, 3*f*
Bis-acrylic composite resin, preformed provisional crowns, 184
Bite wings, vertical, periodontal disease, 24
Black triangles, 59, 59*f*
dento-gingival relationships, 196, 198*f*
Bleaching, 63–66
non-vital bleaching, 65–66, 65*f*, 66*f*
periapical radiographs, 65–66
topically applied systems, 66
whitening strips, 66
stains treated, 64*t*
vital bleaching, 63–65, 64*f*, 65*f*
carbamide peroxide, 63
dentine sensitivity, 65
hydrogen peroxide, 63
urea peroxide, 63
Bleeding, periodontal disease assessment, 22, 23*f*
Bonding materials, onlays, 152*f*, 153, 153*f*
Bony factors, dento-gingival relationships, 196
BPE (basic periodontal examination), 20–21, 20*t*, 22*t*
Bridges *see* Conventional bridges; Fixed prostheses
Brightness *see* Value (brightness)
Bruxism, 69–70
acrylic splints, 70
awake, 69–70
localized occlusal interference splint (LOS), 70, 71*f*
sleep, 69–70
splints, 53
stabilization splints, 70, 70*f*
tooth wear, 49*f*
Buccal preparations
gold crown axial reduction, 110, 110*f*
posterior metal–ceramic crowns, 134–136, 135*f*
Buccal wing retainers, posterior teeth minimal preparation bridges, 234–235
Burs, gold crowns, 106, 106*f*

C

CAD-CAM *see* Computer-aided design and computer-aided manufacture (CAD-CAM)
Canine teeth
conventional bridges, 250, 250*f*
gold crown guidance, 105
teeth colour, 56–57, 57*f*
Cantilever bridges *see* Conventional bridges
CAR (caries adjacent to a restoration), 2
Carbamide peroxide, vital bleaching, 63
Caries *see* Dental caries
Caries adjacent to a restoration (CAR), 2
Cariograms, 10, 10*f*, 11*f*, 12*f*
Castable glass ceramic crowns, 143*f*, 144–145, 145*t*, 146*f*
Casting faults, gold restorations, 120
Cause-related therapy, periodontal disease, 27–28
Cementation
all-ceramic crowns, 149
ceramic veneers *see* Ceramic veneers
gold restorations *see* Gold restorations; Adhesive cements
Centric relation, occlusion, 73
Ceramic
inlays/onlays, 152*f*, 153–154
metal–ceramic crowns, 131, 139
Ceramic crowns *see* All-ceramic crowns
Ceramic posts, 99–100
Ceramic veneers, 155–161
CAD-CAM, 159
cementation, 159–161, 160*f*
isolation, 159–160, 159*f*
multiple veneers, 161, 161*f*
cervical margin, 156, 157*f*
manufacture, 158–159, 158*f*
poorly-aligned teeth, 155, 155*f*, 156*f*
temporization, 158
tooth preparation, 155–158, 156*f*, 157*f*, 158*f*
try-in, 159–161, 159*f*
Cerec machine, CAD-CAM, 148
Cervical chamfer, anterior teeth minimal preparation bridges, 231, 233*f*
Cervical finishing lines
gold crown axial reduction, 110–112, 110*f*
metal–ceramic crowns, 134
Cervical margins, ceramic veneers, 156, 157*f*
C-fibres, dentine pulp complex, 35
Chemical bonding, metal–ceramic crowns, 131
Chemically adhesive luting resins, 125

Chemical treatment, root canal filling removal, 95
Chemomechanical retraction, 173–174
Chroma, Munsell's colour order system, 189*f*, 190
Cingulum test, anterior teeth minimal preparation bridges, 230–231, 231*f*
Circumferential slots and grooves, amalgam cores, 92, 93*f*
Clasps, prosthodontics, 202–203
Clear plastic preformed provisional crowns, 183–184, 184*f*
Clinical assessment, periodontal disease, 20–24
Clinical examination, periodontal disease, 21–24
Cold tests, dental pulp sensitivity, 38*f*, 39
Colour, 189
Colour blindness, 189
Colour changes, tooth wear, 46, 46*f*
Colour vision, 189
Colour vision deficiencies, 189
  tooth shade selection, 192
  see also Colour blindness
Coltène Whaledent (magic foam cord), gingival management (retraction), 173, 173*f*
Combination example, prosthodontics, 204–205, 205*f*, 206*f*
Composites
  cores *see* Cores
  inlays/onlays, 153
Compression bonding, metal–ceramic crowns, 131
Computer-aided design and computer-aided manufacture (CAD-CAM)
  all-ceramic crowns *see* All-ceramic crowns
  ceramic veneers, 159
  conventional bridges, 251, 251*f*
  gold restorations, 119–120, 119*f*, 120*f*
  inlays/onlays, 154
  porcelain jacket crowns, 142
Condensation-cured silicones, impression materials, 169
Condylar inclination, mandible excursive movement, 73, 74*f*
Cone-beam computed tomography, root canal fillings, 42–43
Conformative approach, multiple fixed prostheses, 205–206
Conformative occlusion, gold crowns, 105
Connectors
  conventional bridges, 249, 249*f*
  definition, 239
Consent, metal–ceramic crowns, 132–133
Contact areas, gold restorations, 121, 122
Contact points, metal–ceramic crowns, 139
Conventional bridges, 222, 239–252
  abutment teeth, 240, 240*f*
  aesthetics, 240*f*, 241, 241*f*
  canines, 250, 250*f*
  cantilever bridges, 241–242, 241*f*, 242*f*
    spring, 242
  connectors, 249, 249*f*
  design, 241–246
    CAD-CAM, 251, 251*f*
    examples, 250–251
    see also specific types
  fixed–fixed design, 240–241, 240*f*, 242–243, 242*f*, 243*f*
    tooth preparation, 242–243, 243*f*
  fixed–movable design, 240–241, 243–246, 243*f*, 244*f*
    movable joints, 244–246, 245*f*, 246*f*
    posterior bridges, 244, 245*f*
    stress breaking, 245*f*, 246
  laboratory design, 251
  longevity, 249–250
  occlusion, 249
    Ante's law, 249
  planned, ridge resorption, 220
  pontic design, 247–248
    dome, 247, 247*f*, 248
    hygiene, 247, 247*f*
    modified ridge lap, 247, 247*f*, 248, 248*f*
    ovate, 247, 247*f*, 248
    ridge lap, 247, 247*f*, 248, 248*f*
    saddle, 247, 247*f*, 248, 248*f*
    wash through (sanitary), 247, 247*f*, 248*f*
  premolar teeth, 250–251, 251*f*
  removable prostheses, 250–251
  resin retained bridges *vs.*, 240–241
  shortened dental arch, 250–251
  teeth preparation, 240, 240*f*, 246–247
    full coverage retainers, 246–247
  terminology, 239
Coping materials, all-ceramic crowns, 146, 147*f*
Coping, metal–ceramic crowns, 130
Cores, 87–102
  amalgam, 88, 89–90, 89*f*, 90–92
    circumferential slots and grooves, 92, 93*f*
    dentine bonding, 90–91
    intradental pins, 91–92, 91*f*, 92*f*
    mechanical retention, 91–92
    vital teeth, 90–92
  anterior/single-rooted teeth, 94–100
    active posts, 96
    ceramic posts, 99–100
    fibre posts, 98–99, 98*f*, 99*f*, 100*f*
    metal posts, 96
    non-metal aesthetic posts, 97–100
    passive posts, 97, 97*f*, 98*f*
    posts, 96–100
      see also specific types
    post space creation, 95
    post space preparation, 94–95, 94*f*
    prefabricated passive posts, 97, 98*f*
    pre-tapped posts, 96, 96*f*
    root canal filling removal, 95
    self-threading posts, 96, 96*f*
    weakened roots, 99, 100*f*
    zirconium posts, 99–100
  composite resin, 88–89, 88*f*
    definition, 88
  composites, 90
  definition, 87
  endodontically treated teeth, 93–94
    post placement, 94
    tooth position, 94
    tooth structure, 94
  ferrule effect, 101, 101*f*
  glass ionomer cement (GIC), 89
  materials, 87–90
    see also specific materials
  posterior teeth, 100–101, 101*f*
  resin-modified glass ionomers (RMGIs), 89
  resistance, 101
  retention, 101
  vital teeth, 90–93
    adhesive retention, 90–91
    amalgam, 90–92
    composite resins, 90
    dentine bonding, 90
    elective root canal treatment, 93, 93*f*
    preoperative assessment, 90
Costs, metal–ceramic crowns, 132
Crowns
  all-ceramic *see* All-ceramic crowns
  castable glass ceramic, 143*f*, 144–145, 145*t*, 146*f*
  cementation, dental pulp, 37
  clear plastic preformed provisional, 183–184, 184*f*

Crowns (*Continued*)
custom-formed resin replica *see* Provisional restorations
gold *see* Gold crowns
malleable, 184–185, 185*f*
metal–ceramic *see* Metal–ceramic crowns
polycarbonate, 183, 183*f*
porcelain jacket *see* Porcelain jacket crowns
preformed provisional *see* Provisional restorations
procera, CAD-CAM, 147
three-quarter gold, 112–113, 113*f*, 114*f*
Cusp bevel, gold crown occlusal reduction, 108
Custom-formed resin replica crowns *see* Provisional restorations

D

Dahl concept, tooth wear management, 208
Decalcification, dental pulp, 35*f*
Decontamination, hydrocolloid impression materials, 165
Defence mechanisms, dental pulp, 36
Dehydration, tooth colour, 188, 188*f*
Dental caries, 1–16
detection, 3–5
diagnosis, 3–5
biofilm removal, 3, 3*f*
dental probes, 3–4
DIAGNOdent, 3–4, 6*f*
drying, 3, 4*f*
International Caries Detection and Assessment System (ICDAS II), 3, 5*t*
refraction, 3, 4*f*
emergency treatment, 13
frank cavities, 2
hydroxylapatite crystals, 1
implant retained restorations, 223–224
microcavities, 2
plaque, 1
primary prevention, 1–2
preventative advice, 13
process of, 2*f*
requirements, 2*f*
restoration, 13
risk assessment, 5–12, 6*f*, 13
definition, 5–7
modification, 5–7
re-assessment, 15
risk factors, 7–12, 12*t*
DEPprivation CATEgories (DEPCAT), 7, 7*f*
diet, 8, 9*f*, 12*t*
fluoride, 8–9, 12*t*
oral hygiene, 7–8
past caries experience, 7, 12*t*
plaque, 12*t*
related disease, 12*t*
saliva *see above*
Silness and Löe plaque index, 7–8
social deprivation, 7
Streptococcus mutans, 12*t*
saliva, 9–12, 9*t*, 10*f*, 12*t*
artificial, 9–10
measurement (cariograms), 10, 10*f*, 11*f*, 12*f*
microbiology, 10–12, 14*f*
xerostomia (dry mouth), 9
stabilization, 13
Dental history, periodontal disease *see* Periodontal disease
Dental hygiene, conventional bridges, 247, 247*f*
Dental materials
dental pulp effects, 37
inlays, 153–154
onlays, 153–154
Dental panoramic tomograms (DPTs), periodontal disease, 24, 25*f*
Dental probes
dental caries diagnosis, 3–4
periodontal disease assessment, 23
Dental pulp
crown cementation, 37
decalcification, 35*f*
defence mechanisms, 36
dental material effects, 37
diagnostic procedures, 38–41
see also specific methods
frictional heat, 36–37
health, 36
inflammation treatment, 41
local anaesthetic, 37
marginal microleakage, 37–38, 38*f*
nerves, 34*f*, 35
problems *see* Endodontic problems
restorative procedures, effects of, 36–38
see also specific procedures
sensitivity, 38–40
cold tests, 38*f*, 39
electrical testing, 39–40, 39*f*
heat tests, 39
lingering pain, 40–41, 41*f*
pulp status *vs.*, 40–41
radiography, 40, 40*f*
reliability, 38–39
sharp pain, 40
status, sensitivity *vs.*, 40–41
structure, 34*f*
tooth preparation, 36
vasculature, 35–36
Dentine
chemical changes, 43
derivation, 34
microbial action, 44
production, 34–35
protection, elastomeric impression materials, 170
root canal fillings, 43, 44
sensitivity, vital bleaching, 65
tooth colour, 188
Dentine-bonded crowns *see* All-ceramic crowns
Dentine bonding, 90
amalgam cores, 90–91
tooth wear, 52
Dentine pulp complex, 33–35
Dentinogenesis imperfecta, teeth colour, 58, 58*f*
Dento-facial relationships *see* Aesthetic(s)
Dento-gingival relationships *see* Aesthetic(s)
DEPCAT (DEPprivation CATEgories), dental caries, 7, 7*f*
DEPprivation CATEgories (DEPCAT), dental caries, 7, 7*f*
Depth cuts, gold crown occlusal reduction, 108
Diabetes mellitus, periodontal disease, 19
DIAGNOdent, dental caries diagnosis, 3–4, 6*f*
Diagnostic wax-up, 62, 62*f*
Diastemas, 59, 59*f*
Diet
dental caries, 8, 9*f*, 12*t*
periodontal disease therapy, 28
temporomandibular disorder treatment, 71
tooth wear, 51–52
Diet diaries, 8, 9*f*
Digital photography, tooth shade selection, 192–193, 194*f*
Directly placed composites, inlays, 151–152
Disinfection, impression materials, 176
Dome pontics, 247, 247*f*, 248
DPTs (dental panoramic tomograms), periodontal disease, 24, 25*f*
Dry environments
alginate impression materials, 170
elastomeric impression materials, 170
Drying, dental caries diagnosis, 3, 4*f*
Dry mouth *see* Xerostomia (dry mouth)
Duraphat varnish, dental caries, 8–9

E

Eating disorders, acid erosion, 50, 50*f*
Elastomeric materials *see* Impression materials
Elective root canal treatment, cores, 93, 93*f*
Electrical testing, dental pulp sensitivity, 39–40, 39*f*
Electromagnetic spectrum, 187, 188*f*
Electrosurgery, gingival management (retraction), 174–175, 175*f*
Emergence angle, gold restorations, 121
Endodontic problems, 33–44
  assessment, 41, 42*f*
  cores *see* Cores
  general anatomy, 33–36
  physiology, 33–36
  see also Dental pulp; Dentine pulp complex; Root canal fillings
Erosion, 47
  aetiology, 49–50
  anterior teeth, 48*f*
  appearance, 46, 46*f*
  definition, 45
  eating disorders, 50, 50*f*
  extrinsic acids, 49, 49*t*
  gastro-oesophageal reflux disease (GORD), 50
  intrinsic acids, 50, 50*f*
  molar teeth, 47, 48*f*
Ethyl chloride, pulp sensitivity tests, 39
Eugenol, dentine pulp complex, 37
Expasyl (Kerr Dental), gingival management (retraction), 174
Extraoral examination, occlusion *see* Occlusion
Extrinsic acids, erosion, 49, 49*t*

F

Fabrication, gold restorations *see* Gold restorations
Facebows, semi-adjustable articulators, 78–79, 78*f*, 79*f*, 80*f*
Facial appearance, occlusion, 72, 72*f*
Facial midline, 194, 195*f*
Facial relationships *see* Aesthetic(s)
Family history, periodontal disease, 19
Ferrule effects, cores, 101, 101*f*
Fibre posts, 98–99, 98*f*, 99*f*, 100*f*
Fit surfaces
  dentine-bonded crowns, 144
  gold restorations, 121
  metal–ceramic crowns, 131, 132
Fixed–fixed bridges *see* Conventional bridges
Fixed–movable bridges *see* Conventional bridges
Fixed prostheses, 221–224
  bridges, 222–223
    conventionally prepared, 222
    minimal preparation (resin retained) bridges, 223
  implant retained restorations, 223–224, 225*f*
    dental caries, 223–224
Flowchart guide, tooth shade selection, 192, 192*f*
Fluorescence, 188
Fluoride
  dental caries, 12*t*
  tooth wear, 52
Fluorosis, 188
Fluorosis, teeth colour, 57, 57*f*
Fragility, dentine-bonded crowns, 144
Frank cavities, dental caries, 2
Free-end saddles, removable prostheses, 218–219, 219*f*, 220*f*
Frictional heat, dental pulp, 36–37
Full arch rehabilitation
  multiple fixed prostheses, 205–206
  tooth wear management, 211
Full coverage gold crowns, 114, 114*f*
Fully adjustable articulators, 78
Furcation involvement, periodontal disease assessment, 23*f*, 23*t*, 24*f*, 26

G

Gastro-oesophageal reflux disease (GORD), acid erosion, 50
Generalized tooth wear, 211–213, 211*f*
Genetic disorders, teeth colour, 58
Gingival contour establishment, custom-formed resin replica crowns, 182
Gingival contours
  aesthetic problems, 55–56, 56*f*
  dento-gingival relationships, 198, 199*f*
Gingival management, impression materials *see* Impression materials
Gingival margins
  gold crowns, 106
  metal–ceramic crowns, 139
Gingival retraction, impression materials *see* Impression materials
Gingivitis, periodontal disease, 26
Glass ceramic crowns, 144–147, 145*t*
Glass-infused ceramic copings, 146–147
Glass ionomer cement (GIC), cores, 89
Glass polyalkenoate cement, 124
Gold alloys, 117
  applications, 117
  biocompatibility, 117
  constituents, 117
  metal–ceramic crowns, 130
Gold crowns, 103–116
  axial reduction, 110–112, 112*f*
    approximal preparation, 112, 112*f*, 113*f*
    buccal preparations, 110, 110*f*
    cervical finishing lines, 110–112, 110*f*
    lingual preparations, 110, 110*f*
    'parallel belt', 110, 111, 111*f*
    resistance form, 111
    retention form, 110, 110*f*
    soft tissue protection, 112, 112*f*
  contraindications, 104–105
  examination, 103
  full coverage castings, 114, 114*f*
  impression taking, 114
  indications for, 104
  information sharing, 104
  occlusal reduction, 107–109, 108*f*
    cusp bevel, 108
    depth cuts, 108
    measurement, 108–109, 109*f*
    occlusal offset groove, 106*f*, 109
    retruded contact position-intercuspal position (RCP-ICP), 107
  occlusion recording, 114
  provisional restorations, 114
  targeted history, 103
  three-quarter crowns, 112–113, 113*f*, 114*f*
  tooth preparation, 105–112
    adhesive cements, 107, 107*f*
    axial reduction *see above*
    burs, 106, 106*f*
    gingival margin, 106
    instruments, 106, 106*f*
    occlusal design, 105–106
    occlusal reduction *see above*
    previous restoration removal, 106
    time allocation, 107
Golden proportion, aesthetics, 195, 195*f*, 197*f*
Gold, inlays/onlays, 153, 153*f*
Gold restorations, 117–128
  cementation, 124–126
    chemically adhesive luting resins, 125
    glass polyalkenoate cement, 124
    polycarboxylate cements, 124
    resin luting cements, 125
    resin modified glass polyalkenoate cements, 125

Gold restorations (*Continued*)
restoration luting, 125–126, 125*f*, 126*f*
zinc phosphate cements, 124
checking of, 121–124
refining at bedside, 122–124
refining before appointment, 121, 121*f*
fabrication, 117–120
CAD-CAM, 119–120, 119*f*, 120*f*
casting faults, 120
stone model, 118, 118*f*
tooth preparation, 118, 118*f*
wax patterns, 118–119, 118*f*, 119*f*
supportive phase, 127
Gold veneers, 161, 161*f*
GORD (gastro-oesophageal reflux disease), acid erosion, 50
Group functions, gold crowns, 105
Guide planes, prosthodontics, 201–203, 203*f*, 204*f*

H

Hard tissues
dento-gingival relationships, 196, 197*f*, 198*f*
soft tissue relationships, 198
Heat tests, dental pulp sensitivity, 39
High-speed cutting burs, 175, 175*f*
History taking, periodontal disease, 18–19
Home care, provisional restorations, 179
Horizontal bite wings, periodontal disease, 24
Horizontal lines of face, 195
Horizontal tooth form, 194
Hot-pressed ceramics, 145–146, 145*t*, 146*f*
Hue, Munsell's colour order system, 189, 189*f*
Hydrocolloids, impression materials *see* Impression materials
Hydrogen peroxide, vital bleaching, 63
Hydroxylapatite crystals, dental caries, 1
Hypodontia, 62

I

ICDAS II (International Caries Detection and Assessment System), 3, 5*t*
Immediate prostheses, 221, 221*f*
Immunocompromised patients, periodontal disease, 19
Implants
planned, ridge resorption, 220–221
retained restorations *see* Fixed prostheses
Impression materials, 163–176
agar, 163–164
alginate, 164
dry environments, 170
impression taking, 165
mixing, 165
disinfection, 176
elastomeric materials, 165–170
dentine protection, 170
dry environments, 170
impression checking, 170, 171*f*
undercut blocking out, 169–170, 169*f*
see also specific types
gingival management (retraction), 170–176, 171*f*, 172*f*
aesthetic zone, 172
chemomechanical retraction, 173–174
electrosurgery, 174–175, 175*f*
Expasyl (Kerr Dental), 174
high-speed cutting burs, 175, 175*f*
lasers, 175–176
magic foam cord (Coltène Whaledent), 173, 173*f*
mechanical retraction, 172–173
retraction cord, 172–173, 172*f*, 173–174, 173*f*
surgical retraction, 174–176
hydrocolloids, 163–165
decontamination, 165
impression taking, 164, 164*f*
material setting, 165
see also specific types
silicones, 166–169
addition cured silicones, 166
condensation-cured silicones, 169
packaging, 167, 168*f*
polyethers, 169
trays, 168
two-phase, single-stage technique, 166–168, 166*f*, 167*f*, 168*f*
two-phase, two-stage technique, 168–169, 168*f*
Impression taking
alginate impression materials, 165
gold crowns, 114
Incisor teeth
gold crown guidance, 105
lateral *see* Lateral incisor teeth, missing
openings, dento-gingival relationships, 196
preparations, anterior metal–ceramic crowns, 137, 138*f*
Incisor teeth relationships, 74–75
class I, 75, 75*f*
class II, 75
division I, 75, 76*f*
division II, 75, 76*f*
class III, 75, 76*f*
Incomplete castings, gold restorations, 120
Incorrect dimensions, gold restorations, 120
Indirect restoration shade guides, tooth shade selection, 190–192
Inflammation treatment, dental pulp, 41
Information sharing, gold crowns, 104
Injection-moulded ceramics, all-ceramic crowns, 145–146, 145*t*, 146*f*
Inlays, 151–152, 152*f*
applications, 151–152
definition, 151
directly placed composites, 151–152
materials, 153–154
see also specific materials
tooth preparation, 154, 154*f*, 155*f*
Instrumental shade matching *see* Teeth shade selection
Instruments, gold crowns, 106, 106*f*
Intercuspal positions, occlusion, 72–73, 72*f*, 73*f*
Interdigital fit, gold restorations, 123, 123*f*
International Caries Detection and Assessment System (ICDAS II), 3, 5*t*
International Organization for Standardization (ISO), gold alloys, 117
Interocclusal records, semi-adjustable articulators, 79–82, 81*f*, 82*f*, 83*f*, 84*f*
Intracoronal restorations, posterior teeth minimal preparation bridges, 235, 236*f*, 237*f*
Intradental pins, amalgam cores, 91–92, 91*f*, 92*f*
Intraoral examinations, occlusion *see* Occlusion
Intrinsic acid erosion, 50, 50*f*
Irreversible pulpitis, 93
Irrigants, root canal fillings, 43
ISO (International Organization for Standardization), gold alloys, 117
Isolation, ceramic veneer cementation, 159–160, 159*f*

L

Labial preparations, anterior metal–ceramic crowns, 137, 138*f*
Laboratory-made bridges, 251
  minimal preparation bridges, 228–229
Laboratory preparations, metal–ceramic crowns, 139
Lactobacilli, dental caries, 1, 11*f*, 12*f*
Lamina dura, occlusion, 30
Lasers, gingival management (retraction), 175–176
Lateral excursions, mandible excursive movement, 74, 75*f*, 76, 76*f*, 77*f*
Lateral incisor teeth, missing, 62
  anterior teeth minimal preparation bridges, 232–233, 233*f*
Leucite-reinforced ceramic, CAD-CAM, 149
Light, 187–189
  absorption, 188
  fluorescence, 188
  metamerism, 188, 189*f*
  opalescence, 189
  reflection, 187
  refraction, 188
  scatter, 187
  translucency, 188
  transmission, 188, 188*f*
Lingering pain, dental pulp sensitivity, 40–41, 41*f*
Lingual preparations
  anterior metal–ceramic crowns, 137–139, 138*f*
  gold crown axial reduction, 110, 110*f*
  posterior metal–ceramic crowns, 136, 136*f*
Lingual surfaces, metal–ceramic crowns, 131
Local anaesthetics, dental pulp, 37
Localized occlusal interference splint (LOS), bruxism, 70, 71*f*
Localized tooth wear, 208–209, 208*f*, 209–211, 209*f*, 210*f*, 211*f*
Longevity, conventional bridges, 249–250
Long spans
  posterior teeth minimal preparation bridges, 235, 235*f*, 237*f*
  removable prostheses, 218–219
LOS (localized occlusal interference splint), bruxism, 70, 71*f*
Lower posterior teeth, prosthodontics, 203, 204*f*, 205*f*

M

Machining, all-ceramic crowns, 145*t*
Macromechanical retention, minimal preparation bridges *see* Minimal preparation bridges
Magic foam cord (Coltène Whaledent), gingival management (retraction), 173, 173*f*
Malformed teeth *see* Aesthetic problems
Malleable crowns, preformed provisional crowns, 184–185, 185*f*
Mandible excursive movement, occlusion *see* Occlusion
Marginal fit, gold restorations, 122, 122*f*, 123*f*
Marginal microleakage, dental pulp, 37–38, 38*f*
Margin placement, periodontal considerations, 29
Maryland bridge, 229
Mastication, missing teeth replacement, 216–217
Mechanical interlocking, metal–ceramic crowns, 131
Mechanical retention, amalgam cores, 91–92
Mechanical retraction, gingival management (retraction), 172–173
Mechanical tooth wear, 46–47, 47*f*
Medical history, periodontal disease *see* Periodontal disease
Metal–ceramic bonds, metal–ceramic crowns, 130–131
Metal–ceramic crowns, 129–140, 129–132
  advantages, 131
  anterior, 137–139
    approximal preparation, 139, 139*f*
    incisal preparation, 137, 138*f*
    labial preparation, 137, 138*f*
    palatal/lingual preparation, 137–139, 138*f*
  axial reduction, 134
  ceramic, 131, 139
  cervical finishing lines, 134
  design, 139
  disadvantages, 131–132
  indications, 129
  laboratory preparation, 139
  metal–ceramic bond, 130–131
  metals, 130, 139
  occlusal reduction, 133
  patient consent, 132–133
  posterior, 134–137
    approximal preparation, 136–137, 136*f*, 137*f*
    buccal preparation, 134–136, 135*f*
    lingual preparation, 136, 136*f*
    occlusal preparation, 134
  preparation finishing, 134
  resistance form, 134
  retention form, 134
  structure, 129, 130*f*
  tooth preparation, 133–134
  value, 190, 190*f*
  wax-up, 133, 133*f*
Metal coverage, metal–ceramic crowns, 139
Metal(s), metal–ceramic crowns, 130, 139
Metal posts, 96
Metal preformed provisional crowns, 184, 184*f*
Metamerism, 188, 189*f*
Michigan splints, 53, 53*f*, 70, 70*f*
Microbial leakage, root canal fillings, 44
Microbiology, dental caries, 10–12, 14*f*
Microcavities, dental caries, 2
Microcracks, root canal fillings, 43, 44*f*
Microleakage, 2, 3*f*
  marginal, dental pulp, 37–38, 38*f*
Midline, dento-facial relationships, 195, 195*f*
Miller's index, tooth mobility, 24*t*
Minimal preparation bridges, 223, 227–238, 228*f*
  anterior teeth, 230–233, 230*f*, 232*f*
    cervical chamfer, 231, 233*f*
    cingulum test, 230–231, 231*f*
    missing lateral incisors, 232–233, 233*f*
    occlusal clearance, 231–232
    orthodontic relapse prevention, 233, 234*f*
  applications, 227
  conventional bridges *vs.*, 240–241
  definition, 227
  history, 227–230
  macromechanical retention, 228–229
    adhesive resin luting cements, 229–230, 230*f*
    laboratory-made bridges, 228–229
    natural tooth pontic, 228, 228*f*, 229*f*
    Rochette bridges, 228–229, 229*f*
    solid wing retainers, 229–230
  posterior teeth, 234–238, 234*f*, 235*f*
    buccal wing retainers, 234–235
    intracoronal restorations, 235, 236*f*, 237*f*
    long span, 235, 235*f*, 237*f*
Missing teeth, provisional restorations, 185, 185*f*, 186*f*

Missing teeth replacement, 215–226
fixed prostheses *see* Fixed prostheses
indications, 215–217
appearance, 215, 216*f*
mastication, 216–217
occlusal stability, 215–216, 216*f*
psychology, 217
quality of life, 217
speech, 217
methods, 217–226
see also specific methods
necessity for, 215
orthodontic treatment, 224–226, 225*f*
removable prostheses *see* Removable prostheses
Mixing, alginate impression materials, 165
Model mounting, semi-adjustable articulators, 78–82
Modified ridge lap pontics, 247, 247*f*, 248, 248*f*
Moisture contamination, composite resin cores, 88
Molar teeth
abrasion, 47
attrition, 47–49, 48*f*
erosion, 47, 48*f*
tooth wear, 47–49, 48*f*
Mouth, dry *see* Xerostomia (dry mouth)
Movable joints, fixed–movable bridges, 244–246, 245*f*, 246*f*
Multiple edentulous spaces, removable prostheses, 218–219, 218*f*
Multiple fixed prostheses, 205–206
conformative approach, 205–206
full arch rehabilitation, 205–206
Multiple retained roots, prosthodontics, 205, 206*f*
Multiple veneers, cementation, 161, 161*f*
Munsell's colour order system, 189–190, 189*f*
chroma, 189*f*, 190
hue, 189, 189*f*
value, 189*f*, 190

N

Natural tooth pontic, minimal preparation bridges, 228, 228*f*, 229*f*
Nerves, dental pulp, 34*f*, 35
Neurogenic inflammation, dentine pulp complex, 35
Nobel Biocare, CAD-CAM, 147, 147*f*
Non-metal aesthetic posts, 97–100
Non-steroidal anti-inflammatory drugs (NSAIDs), temporomandibular disorder treatment, 71
Non-vital bleaching *see* Bleaching

O

Occlusal clearance
anterior teeth minimal preparation bridges, 231–232
tooth wear restorative management, 208–213
Occlusal design, gold crowns, 105–106
Occlusal factors, metal–ceramic crowns, 139
Occlusal forces, root canal fillings, 43
Occlusal offset groove, gold crowns, 106*f*, 109
Occlusal preparation, posterior metal–ceramic crowns, 134
Occlusal reduction
gold crowns *see* Gold crowns
metal–ceramic crowns, 133
rest seats, 202, 202*f*
Occlusal stability, missing teeth replacement, 215–216, 216*f*
Occlusal surfaces, metal–ceramic crowns, 131
Occlusal table
gold crowns, 105
gold restorations, 121
Occlusal vertical dimension (OVD), 196
Occlusion, 69–86
checking, gold restorations, 123, 123*f*
conventional bridges *see* Conventional bridges
custom-formed resin replica crowns, 181–182, 182*f*
definition, 69
extraoral examination, 72
facial appearance, 72, 72*f*
temporomandibular joints, 72
intraoral examination, 72–76
centric relation, 73
intercuspal position, 72–73, 72*f*, 73*f*
mandible excursive movement *see below*
retruded contact position (RCP), 73
terminal hinge axis position, 73, 73*f*
lamina dura, 30
management, 76–77
see also Articulators
mandible excursive movement, 73–76
anterior guidance, 74–76
condylar inclination, 73, 74*f*
lateral excursion, 74, 75*f*, 76, 76*f*, 77*f*
posterior guidance, 73–74
protrusion, 73, 74–75, 74*f*
see also Incisor teeth relationships
patient examination, 72–76
see also specific types
periodontium, 30
recording, gold crowns, 114
reorganized, gold crowns, 105
see also Bruxism; Temporomandibular disorders (TMD)
Odontoblasts, 34
damage to, 34–35
Onlays, 152–153, 152*f*, 153*f*
bonding materials, 152*f*, 153, 153*f*
definition, 151
dental materials, 153–154
see also specific materials
indications, 152
tooth preparation, 154, 154*f*, 155*f*
Opalescence, 189
Oral hygiene
assessment, periodontal disease assessment, 22
dental caries, 7–8
periodontal disease therapy, 26
provisional restorations, 177
Orthodontic treatment
missing teeth replacement, 224–226, 225*f*
relapse prevention, 233, 234*f*
Ovate pontics, 247, 247*f*, 248
OVD (occlusal vertical dimension), 196
Overdentures, tooth wear restorative management, 208, 208*f*
Overeruption, elective root canal treatment, 93, 93*f*
Oxide layers, metal–ceramic crowns, 130

P

Packaging, silicone impression materials, 167, 168*f*
Pain
dental pulp sensitivity, 40–41, 41*f*
lingering, 40–41, 41*f*
sharp, 40
Pain relief, periodontal disease, 27
Palatal preparation, anterior metal–ceramic crowns, 137–139, 138*f*

Pantographs, interocclusal record, 83, 84*f*, 85*f*
'Parallel belt,' gold crown axial reduction, 110, 111, 111*f*
Partial dentures *see* Prosthodontics
Particle size, composite resin cores, 88
Passive posts, 97, 97*f*, 98*f*
Past caries experience, 7, 12*t*
Patient compliance, periodontal disease, 19
Peg laterals, 61
Periapical radiographs
  non-vital bleaching, 65–66
  periodontal disease, 24, 25*f*
Perimolysis, 46, 46*f*
Periodontal bone loss, 24
Periodontal disease, 17–32
  assessment, 19–26
    attachment loss, 23, 23*f*
    basic periodontal examination (BPE), 20–21, 20*t*, 22*t*
    bleeding on probing, 22, 23*f*
    clinical assessment, 20–24
    clinical examination, 21–24
    furcation involvement, 23*f*, 23*t*, 24*f*, 26
    oral hygiene assessment, 22
    plaque charts, 22, 22*f*
    probing depth, 23
    radiographic examination *see below*
    tooth mobility, 24*t*, 26
  dental history, 18–19
    appliances, 20
    patient compliance, 19
    plaque control, 20
    previous history, 22
  diagnosis, 26–27
    gingivitis, 26
    periodontal stability, 26
  family history, 19
  history taking, 18–19
  medical history, 18
    diabetes mellitus, 19
    immunocompromised patients, 19
  periodontitis, 26–27
    aggressive, 25*f*, 26–27, 27*f*
  radiographic examination, 24–26
    report, 25–26
    types, 24–25, 25*f*
  risk concept, 18
  social history, 19
    alcohol consumption, 19
    smoking, 19
    stress, 19
  susceptibility to, 17–18
  treatment planning, 27–29
    phase 1 (pain relief), 27
    phase 2 (cause-related therapy), 27–28
    phase 3 (re-examination), 28–29
    phase 4 (definitive treatment), 29
    phase 5 (maintenance/supportive periodontal therapy), 29
Periodontal stability, 26
Periodontitis
  aggressive, 25*f*, 26–27, 27*f*
  periodontal disease *see* Periodontal disease
Periodontium, occlusion, 30
Periradicular radiolucency, root canal fillings, 42
Photographic mirrors, gold crowns, 106, 106*f*
Pitting, gold restorations, 120
Plaque
  dental caries, 1, 12*t*
  periodontal disease, 20
  removable prostheses, 217, 218*f*
Plaque charts, 22, 22*f*
Plastic restoration failure, gold crown, 104, 104*f*
Polishing, periodontal disease therapy, 28
Polycarbonate crowns, 183, 183*f*
Polycarboxylate cements, 124
Polyethers, impression materials, 169
Pontics
  conventional bridges *see* Conventional bridges
  definition, 239
Poorly-aligned teeth, ceramic veneers, 155, 155*f*, 156*f*
Porcelain jacket crowns, 141–143, 142*f*
  aesthetic restoration, 142, 142*f*
  CAD-CAM, 142
  sintering, 142–143, 145*t*
  tooth preparation, 142, 142*f*
Porosity, gold restorations, 120
Posterior fixed–movable bridges, 244, 245*f*
Posterior guidance, mandible excursive movement, 73–74
Posterior teeth
  cores, 100–101, 101*f*
  lower, prosthodontics, 203, 204*f*, 205*f*
  metal–ceramic crowns *see* Metal–ceramic crowns
  minimal preparation bridges *see* Minimal preparation bridges
Posts, 96–100
  active, 96
  ceramic, 99–100
  fibre, 98–99, 98*f*, 99*f*, 100*f*
  metal, 96
  non-metal aesthetic, 97–100
  passive, 97, 97*f*, 98*f*
  placement, endodontically treated teeth, 94
  prefabricated passive, 97, 98*f*
  pre-tapped, 96, 96*f*
  self-threading, 96, 96*f*
  space creation, 95
  space preparation, 94–95, 94*f*
  zirconium, 99–100
Precision attachments *see* Prosthodontics
Prefabricated passive posts, 97, 98*f*
Preformed provisional crowns *see* Provisional restorations
Premolar teeth, conventional bridges, 250–251, 251*f*
Preparation finishing, metal–ceramic crowns, 134
Pre-tapped posts, 96, 96*f*
Preventative advice, dental caries, 13
Previous restoration removal, gold crowns, 106
Primary prevention, dental caries, 1–2
Probing depth, periodontal disease assessment, 23
Procera crowns, CAD-CAM, 147
Prostheses
  immediate, 221, 221*f*
  transitional, 221
Prostheses, removable *see* Removable prostheses
Prosthodontics, 201–214
  clasps, 202–203
  guide planes, 201–203, 203*f*, 204*f*
  precision attachments, 203–205
    combination example, 204–205, 205*f*, 206*f*
    lower posterior teeth, 203, 204*f*, 205*f*
    multiple retained roots, 205, 206*f*
  rest seats, 201–203, 202*f*
    occlusal reduction, 202, 202*f*
  shortened dental arch, 205
  tooth wear restorative management, 209, 209*f*, 210*f*
  undercuts, 201–203
  see also Multiple fixed prostheses
Protrusion, mandible excursive movement, 73, 74–75, 74*f*
Provisional restorations, 177–186, 178*f*
  advice, 185–186
  aesthetics, 178
  bacterial leakage prevention, 178
  characteristics, 178–179

Provisional restorations (*Continued*)
custom-formed resin replica crowns, 179–182, 179*f*
aesthetics, 181–182
gingival contour establishment, 182
manufacture, 179–181, 179*f*, 180*f*
modifications, 181, 181*f*
occlusion, 181–182, 182*f*
gold crowns, 114
home care, 179
missing teeth, 185, 185*f*, 186*f*
oral health care, 177
preformed provisional crowns, 183–185
bis-acrylic composite resin, 184
clear plastic crowns, 183–184, 184*f*
malleable crowns, 184–185, 185*f*
metal, 184, 184*f*
original crown modification, 184, 184*f*
polycarbonate crowns, 183, 183*f*
WAMkey, 184, 185*f*
tooth preparation adequacy, 178
tooth sensitivity prevention, 178
Psychology, missing teeth replacement, 217
Pulp *see* Dental pulp
Pulpitis
acute, 38
irreversible, 93

Q

Quality of life, missing teeth replacement, 217

R

Radiography
dental pulp sensitivity, 40, 40*f*
periapical *see* Periapical radiographs
periodontal disease *see* Periodontal disease
root canal fillings, 42–43, 43*f*
RCP-ICP (retruded contact position-intercuspal position), gold crown occlusal reduction, 107
RCP (retruded contact position), occlusion, 73
Re-examination, periodontal disease, 28–29
Reflection, 187
Refraction, 188
dental caries diagnosis, 3, 4*f*
Related disease, dental caries, 12*t*
Remineralization, white spot lesions, 1, 2*f*
Removable prostheses, 217–221
conventional bridges, 250–251
free-end saddles, 218–219, 219*f*, 220*f*
immediate prostheses, 221, 221*f*
long spans, 218–219
multiple edentulous spaces, 218–219, 218*f*
plaque accumulation, 217, 218*f*
ridge resorption, 219–221, 220*f*
planned bridges, 220
planned implants, 220–221
severe alveolar bone loss, 221, 222*f*
single missing teeth with spacing, 221, 223*f*, 224*f*
transitional prostheses, 221
Reorganized occlusion, gold crowns, 105
Resin luting cements
all-ceramic crowns, 149
gold restoration cementation, 125
Resin-modified glass ionomers (RMGIs), cores, 89
Resin modified glass polyalkenoate cements, gold restoration cementation, 125
Resin retained bridges *see* Minimal preparation bridges
Resistance, cores, 101
Resistance form
gold crown axial reduction, 111
metal–ceramic crowns, 134
Restoration
advanced, periodontal considerations, 30, 30*f*
dental caries, 13
provisional *see* Provisional restorations
tooth wear, 53
Restoration luting, gold restoration cementation, 125–126, 125*f*, 126*f*
Restorative cycle, 2–3
Rest seats, prosthodontics *see* Prosthodontics
Retainers, 239
Retention, cores, 101
Retention form
gold crown axial reduction, 110, 110*f*
metal–ceramic crowns, 134
Retraction cord, gingival management (retraction), 172–173, 172*f*, 173–174, 173*f*
Retruded contact position-intercuspal position (RCP-ICP), gold crown occlusal reduction, 107
Retruded contact position (RCP), occlusion, 73
Ridge lap pontics, 247, 247*f*, 248, 248*f*
Ridge resorption, removable prostheses *see* Removable prostheses
RMGIs (resin-modified glass ionomers), cores, 89
Rochette bridges, minimal preparation bridges, 228–229, 229*f*
Root canal fillings
radiography, 42–43, 43*f*
removal, cores, 95
survival of, 43–44
Root surface instrumentation (RSI), periodontal disease therapy, 28
RSI (root surface instrumentation), periodontal disease therapy, 28
Rubber dam isolation, composite resin cores, 88

S

Saddle pontics, 247, 247*f*, 248, 248*f*
Saliva
artificial, 9–10
dental caries *see* Dental caries
tooth wear, 52
Sanitary pontics, 247, 247*f*, 248*f*
Scaling, periodontal disease therapy, 28
Scatter, 187
SDA *see* Shortened dental arch (SDA)
Self-threading posts, 96, 96*f*
Semi-adjustable articulators *see* Articulators
Sensitivity, dental pulp *see* Dental pulp
Severe alveolar bone loss, removable prostheses, 221, 222*f*
Shade-matching devices, 192, 193*f*
Sharp pain, dental pulp sensitivity, 40
Shortened dental arch (SDA), 216
conventional bridges, 250–251
prosthodontics, 205
temporomandibular dysfunction, 216
tooth wear, 216–217, 217*f*
Shoulders, dentine-bonded crowns, 143, 143*f*, 144*f*
Silicones, impression materials *see* Impression materials
Silness and Löe plaque index, 7–8
Simple hinge articulators, 77, 77*f*
Single missing teeth with spacing, removable prostheses, 221, 223*f*, 224*f*
Single-rooted teeth, cores *see* Cores
Sintering, porcelain jacket crowns, 142–143, 145*t*
Sleep bruxism, 69–70
Smile line
definition, 55
dento-facial relationships, 195–196

Smoking, periodontal disease, 19, 28
Smoothing, metal–ceramic crowns, 130
Social history
  dental caries, 7
  periodontal disease *see* Periodontal disease
Soft tissues
  dento-gingival relationships, 197–198
  hard tissue relationships, 198
  protection, gold crown axial reduction, 112, 112*f*
Solid wing retainers, minimal preparation bridges, 229–230
Span, 239
Speech, missing teeth replacement, 217
Splints
  bruxism, 53
  tooth wear, 52–53
  see also Stabilization splints
Spring cantilever bridges, 242
SPT (supportive periodontal therapy), periodontal disease, 29
Stabilization splints
  bruxism, 70, 70*f*
  temporomandibular disorder treatment, 71
Stone model, gold restorations, 118, 118*f*
Strength, metal–ceramic crowns, 131, 132
Streptococcus mutans, dental caries, 1, 11*f*, 12*f*, 12*t*
Stress breaking, fixed–movable bridges, 245*f*, 246
Stress, periodontal disease, 19
Structural integrity, root canal fillings, 43
Substructure design, metal–ceramic crowns, 132, 132*f*
Supportive periodontal therapy (SPT), periodontal disease, 29
Supportive phase, gold restorations, 127
Surgical crown lengthening, tooth wear restorative management, 207–208, 207*f*, 208*f*
Surgical retraction, gingival management (retraction), 174–176
Systemic sclerosis, gold crown contraindication, 104–105

T

Targeted history, gold crowns, 103
Teeth
  anterior *see* Anterior teeth
  colour *see* Teeth colour
  extraction, periodontal disease therapy, 28
  fracture, gold crown, 104, 104*f*
  malformed *see* Aesthetic problems
  missing *see under* missing teeth
  mobility *see* Teeth mobility
  poorly-aligned, ceramic veneers, 155, 155*f*, 156*f*
  position, cores, 94
  posterior *see* Molar teeth; Posterior teeth
  preparation *see* Teeth preparation
  sensitivity prevention, provisional restorations, 178
  shape problems *see* Aesthetic problems
  single-rooted, cores *see* Cores
  size, dento-gingival relationships, 196, 198*f*
  size problems *see* Aesthetic problems
  spacing *see* Aesthetic problems
  structure, cores, 94
  wear *see* Teeth wear
Teeth colour
  correction *see* Aesthetic problems
  dehydration, 188, 188*f*
  dentine, 188
  dento-gingival relationships, 196, 198*f*
Teeth mobility
  Miller's index, 24*t*
  occlusion, 29
  periodontal disease assessment, 24*t*, 26
Teeth preparation
  adequacy, provisional restorations, 178
  ceramic veneers, 155–158, 156*f*, 157*f*, 158*f*
  conventional bridges *see* Conventional bridges
  dental pulp, 36
  fixed–fixed bridges, 242–243, 243*f*
  gold crowns *see* Gold crowns
  gold restorations, 118, 118*f*
  inlays, 154, 154*f*, 155*f*
  metal–ceramic crowns, 131, 133–134
  onlays, 154, 154*f*, 155*f*
  porcelain jacket crowns, 142, 142*f*
Teeth shade selection, 190–193
  instrumental shade matching, 192–193
    digital photography, 192–193, 194*f*
    shade-matching devices, 192, 193*f*
  traditional shade matching, 190–192
    colour vision deficiencies, 192
    flowchart guide, 192, 192*f*
    indirect restoration shade guides, 190–192
    value, 190, 190*f*
    VITA classical shades, 190, 191*f*, 192*f*
Teeth wear, 45–54, 206–213
  acidic challenges, 47
  aetiology, 49–50
  anterior teeth, 46–47, 46*f*, 47*f*, 48*f*
  bruxism, 49*f*
  clinical appearance, 46–49
  detection, 50–51, 51*f*
  management, 50–53
  mechanical, 46–47, 47*f*
  molars, 47–49, 48*f*
  monitoring, 50–51, 52*f*
  preventative management, 51–53
    dentine bonding agents, 52
    diet, 51–52
    fluoride, 52
    restoration, 53
    saliva, 52
    splint therapy, 52–53
  restorative management, 207–213, 207*f*
    Dahl concept, 208
    full arch rehabilitation, 211
    generalized tooth wear, 211–213, 211*f*
    localized tooth wear, 208–209, 208*f*, 209–211, 209*f*, 210*f*, 211*f*
    occlusal clearance, 208–213
    overdentures, 208, 208*f*
    partial dentures, 209, 209*f*, 210*f*
    surgical crown lengthening, 207–208, 207*f*, 208*f*
  sensitivity, 49
  shortened dental arch (SDA), 216–217, 217*f*
  see also specific types
Temporary cover, periodontal considerations, 29
Temporization, ceramic veneers, 158
Temporomandibular disorders (TMD), 70–72
  aetiology, 70–71
  definition, 70–71, 71*f*
  epidemiology, 71
  treatment, 71
Temporomandibular dysfunction, shortened dental arch (SDA), 216
Temporomandibular joints, occlusion, 72

Terminal hinge axis position, occlusion, 73, 73*f*
Tetracycline staining, 57, 57*f*
Thermal treatment, root canal filling removal, 95
3M ESPE, CAD-CAM, 147, 148*f*
Three-quarter gold crowns, 112–113, 113*f*, 114*f*
Time allocation, gold crowns, 107
Titanium alloys, metal–ceramic crowns, 130
TMD *see* Temporomandibular disorders (TMD)
TOC (total occlusal convergence), 110, 110*f*
Topically applied systems, non-vital bleaching, 66
Total occlusal convergence (TOC), 110, 110*f*
Traditional shade matching *see* Teeth shade selection
Transitional prostheses, 221
Translucency, 188
Transmission, 188, 188*f*
Trauma, malformed teeth, 61
Trays, silicone impression materials, 168
Try-in, ceramic veneers, 159–161, 159*f*
Two-phase, single-stage technique, silicone impression materials, 166–168, 166*f*, 167*f*, 168*f*
Two-phase, two-stage technique, silicone impression materials, 168–169, 168*f*

U

Undercuts
  blocking out, 169–170, 169*f*
  prosthodontics, 201–203
Urea peroxide, vital bleaching, 63

V

Value (brightness)
  metal–ceramic crowns, 132, 132*f*, 190, 190*f*
  Munsell's colour order system, 189*f*, 190
  tooth shade selection, 190, 190*f*
Van der Waals forces, metal–ceramic crowns, 131
Vasculature, dental pulp, 35–36
Veneers, multiple, 161, 161*f*
Vertical bite wings, periodontal disease, 24
Vertical dimensions, dento-facial relationships, 196
Vertical tooth form, aesthetics, 194
VITA classical shades, tooth shade selection, 190, 191*f*, 192*f*
VITA In-Ceram, CAD-CAM, 147
Vital bleaching *see* Bleaching
Vital teeth, cores *see* Cores

W

WAMkey, 184, 185*f*
Wash through pontics, 247, 247*f*, 248*f*
Wax patterns
  gold restorations, 118–119, 118*f*, 119*f*
  metal–ceramic crowns, 133, 133*f*
Whitening strips, 66
White spot lesions, remineralization, 1, 2*f*
WHO probe, basic periodontal examination (BPE), 19, 22*f*

X

Xerostomia (dry mouth)
  dental caries, 9
  gold crown contraindication, 104–105

Z

Z100 composite, CAD-CAM, 149
Zinc phosphate cements, gold restorations, 124
Zirconium posts, 99–100